“ A Comprehensive Review on Battery Thermal Management System for Electric Vehicles “

~ INDEX ~

INTRODUCTION

Objective : To explain why thermal management systems is crucial for batteries especially for electric vehicles. So starting from the energy storage of a battery, with the growing use of renewable energy, managing the inconsistent supply from sources like wind and solar power has become a challenge for energy grids. Energy storage systems (ESS) like lithium-ion batteries (LIB) and reversible fuel cells (RFC) can help by storing excess energy when supply is high and releasing it when demand exceeds supply. Battery Thermal Management Systems (BTMS) play a crucial role in battery storage, especially for electric vehicles (EVs) and other high-performance applications like

Temperature Regulation: BTMS maintain optimal battery temperatures, ensuring safety, efficiency, and longevity. This is vital because temperature fluctuations can significantly impact battery performance. When the overall temperature of battery reaches 50°C the cell becomes unstable due to change in internal chemistry as the reactions are temperature dependent resulting in degrading performance, degraded electrode and battery life. Hence BTMS is used to control this.

Preventing Thermal Runaway: By managing heat generation during charging and discharging, BTMS prevent thermal runaway, which can lead to battery failure or even explosions. Electric vehicles (EVs) are becoming more popular because they are better for the environment and cost less to operate. However, the lithium-ion batteries (LIBs) in EVs need to stay at the right temperature to work well and stay safe. Liquid-cooled battery thermal management systems (LC-BTMS) are very effective at keeping these batteries cool, so they are used in many EVs.

Enhancing Performance: Proper thermal management allows batteries to operate at their best, providing consistent power output and enabling fast charging capabilities. With the rapid development of new energy vehicle the mileage and the battery is expected to bee higher and higher. So this makes us to increase the charging speed of battery as the data of charging of the battery has always been a paint point for the development of the new energy vehicle.

Extending Battery Life: Keeping batteries within their ideal temperature range reduces wear and tear, thereby extending their lifespan. Lithium-ion (Li-ion) batteries are widely used in electric vehicles and other applications due to their high specific energy, high energy density, and efficiency. However, they generate significant heat at high charge and discharge rates, potentially leading to thermal runaway. To enhance their lifespan and performance, a Battery Thermal Management System (BTMS) is essential.

Improving Energy Efficiency: Efficient thermal management minimizes energy losses and maximizes the overall efficiency of the battery system.

Role of BTMS in EVs :

Temperature Control

BTMS keep the battery at the right temperature. This is key because if the battery gets too hot or too cold, it won't work as efficiently. Whether you're driving in extreme heat or cold, BTMS ensure the battery performs well.

Safety First

BTMS prevent what's known as thermal runaway. This is when the battery's temperature spikes uncontrollably, potentially causing fires or explosions. By managing the heat, BTMS keep things safe.

Top Performance

With BTMS, the battery can deliver consistent power. This means your EV runs smoothly, whether you're just running errands or on a long road trip. Plus, BTMS support fast charging, which is super convenient.

Longer Battery Life

By keeping the battery within a stable temperature range, BTMS reduce wear and tear. This helps the battery last longer, saving you money in the long run since you won't need to replace it as often.

Energy Efficiency

Good thermal management means less energy is wasted on heating or cooling the battery. This boosts the overall efficiency of the vehicle and can even improve your driving range.

Integrated System

Modern BTMS work together with other systems in your EV to manage heat in real-time. They use smart algorithms to predict thermal loads and adjust cooling methods on the fly.

There are different ways to manage battery temperature, like liquid cooling, air cooling, and using phase change materials (PCMs). Each method has its pros and cons, and the best choice depends on the specific needs of the vehicle.

The global transition towards clean energy solutions is driving innovations in electric vehicles (EVs), as traditional combustion engines are gradually being replaced by hybrid and fully electric systems. Lithium-ion (Li-ion) batteries are preferred for these applications due to their high energy and power densities. However, these batteries are highly sensitive to operating temperatures, particularly under high charging and discharging rates, referred to as C-rates. With the increasing adoption of electric vehicles (EVs) to combat global warming and pollution, efficient thermal management systems are crucial for ensuring the safety, performance and longevity of lithium ion batteries, which are sensitive to temperature variation.

BACKGROUND THEORY/BASIC CONCEPTS

Basic Components of a Li-ion Battery

1. Anode (Negative Electrode):
    - Typically made of graphite.
    - During discharge, the lithium ions move from the anode to the cathode through the electrolyte.
    - During charging, they move back from the cathode to the anode.
2. Cathode (Positive Electrode):
    - Made from a lithium metal oxide (such as lithium cobalt oxide, lithium iron phosphate, or lithium manganese oxide).
    - Stores the lithium ions when the battery is charged.
3. Electrolyte:
    - A chemical medium that allows the flow of lithium ions between the cathode and anode.
    - Usually consists of a lithium salt dissolved in an organic solvent.
4. Separator:
    - A thin, porous membrane that physically separates the anode and cathode.
    - Allows the flow of ions but prevents electrical contact between the electrodes to avoid short circuits.
5. Current Collectors:
    - Typically made of copper for the anode and aluminum for the cathode.
    - Collect and transport electrons from the anode and cathode to the external circuit.

How It Works

- Charging:
    - When the battery charges, lithium ions move from the cathode through the electrolyte to the anode.
    - Electrons flow through the external circuit to balance the charge.
- Discharging:
    - During discharge, lithium ions move back from the anode to the cathode, releasing energy.
    - Electrons flow through the external circuit, providing power to connected devices.

This structure allows for high energy density, efficiency, and a relatively low self-discharge rate, making Li-ion batteries ideal for a wide range of applications, from smartphones to electric vehicles.

What can be the possible disadvantages of a Lithium Ion battery being used in Electric Vehicles (EVs). Using Li-ion batteries directly in electric vehicles (EVs) without a Battery Thermal Management System (BTMS) comes with several disadvantages:

Overheating

Without proper cooling, Li-ion batteries can overheat, especially during high-rate charging and discharging. This can lead to reduced efficiency and performance.

Thermal Runaway

Overheating can lead to thermal runaway, a dangerous condition where the battery's temperature increases uncontrollably, potentially causing fires or explosions.

Reduced Battery Life

High temperatures accelerate the degradation of battery cells, shortening their lifespan and necessitating more frequent replacements, which can be costly.

Performance Issues

Without temperature regulation, the battery's performance can fluctuate, leading to inconsistent power output. This affects the vehicle's reliability and driving range.

Safety Risks

In the absence of BTMS, the risk of safety incidents like fires or explosions increases significantly, posing a threat to the vehicle and its occupants.

Lower Efficiency

Overheating leads to energy loss, reducing the overall efficiency of the EV. A well-regulated temperature helps maintain optimal energy efficiency.

Limited Fast Charging

Without BTMS, fast charging can generate excessive heat, making it impractical and unsafe to use, thus limiting the convenience for EV users.

In summary, a BTMS is essential for maintaining safety, performance, and efficiency in electric vehicles. It helps manage the heat generated by Li-ion batteries, ensuring they operate within safe and optimal temperature ranges.

Current standards or expectations of battery performance in the EV industry are as follows :

Performance Standards

1. **Energy Density**: High energy density is essential for longer driving ranges. EV batteries typically aim for an energy density of around 150-250 Wh/kg.
2. **Power Density**: This determines how quickly a battery can deliver energy. Higher power density is important for better acceleration and performance.
3. **Cycle Life**: The number of charge-discharge cycles a battery can undergo before its capacity drops to a certain level (usually 80%). EV batteries are expected to have a cycle life of around 1,000-2,000 cycles.

4. **Efficiency**: The ratio of energy output to energy input. Higher efficiency means less energy is lost during charging and discharging.

5. **Safety**: Compliance with safety standards to prevent thermal runaway, fires, and other hazards. This includes robust Battery Management Systems (BMS) and thermal management systems (BTMS).

Durability and Reliability

1. **Temperature Tolerance**: Batteries should perform well across a wide range of temperatures, from extreme cold to high heat.

2. **Degradation Rate**: Minimizing capacity loss over time and usage. Slower degradation rates are preferred for longer-lasting batteries.

3. **Consistency**: Ensuring consistent performance over the battery's lifespan, without significant drops in capacity or efficiency.

Environmental and Sustainability Standards

1. **Eco-Friendly Materials**: Use of materials that are less harmful to the environment and easier to recycle.

2. **Recyclability**: Batteries should be designed for easy disassembly and recycling at the end of their life cycle.

3. **Lifecycle Analysis**: Assessing the environmental impact of the battery from production to disposal.

Why we use Lithium Ion Batteries LIBs :

Among these, EVs have gained significantly popularity, with over 10 million sold world wide in 2022 and a projected market growth to 228 million by2030. Lithium-ion batteries have become the preferred energy storage solutions for EVs due to their high energy density, with the potential to provide up to 350Wh/kg, and cost-effectiveness, reaching a production cost of 100 dollars/kWh.

LIBs have gone continuous material development, with various chemistries being explored

MAIN TECHNIQUES FOR BTMS

Objective : This section reviews the various cooling techniques used for battery thermal management systems. This section also compares different methods, their advantages, disadvantages and highlights example of each.

Why to focus on cooling system of battery :

Fast Charging Purpose : For the application of fast charging mechanismwe should have a good cooling system which then comes on the design part of the battery as how it should be designed to get the proper amount and the type of cooling to support the fast charging mechanism.

High temperature can accelerate side reactions, leading to capacity loss and potential fire hazards, while low temperatures reduce discharge capacity and may cause lithium dendrite growth

Non uniform temperature further exacerbate battery performance and life span

Air Cooling : An air cooling system in a Battery Thermal Management System (BTMS) uses airflow to regulate the temperature of lithium-ion batteries. This method involves directing cool air over the battery cells, either naturally or using fans (forced air cooling). It's a relatively simple and cost-effective approach, effective for moderate temperature management. Air cooling helps dissipate heat generated during charging and discharging cycles, preventing overheating and ensuring the battery operates within its optimal temperature range. However, its efficiency can be limited in high-power applications where substantial heat is generated, making it less effective compared to liquid cooling systems in those scenarios.

Air cooling offers simplicity and reliability but struggles with temperature uniformity.

Which includes about sandwiched heat pipe cooling systems SHCS :

study compares four cooling methods: natural convection, SHCS with natural convection, SHCS with forced convection, and forced convection for both SHCS and the battery. The SHCS is proven to significantly reduce maximum battery temperature compared to natural convection alone. Using forced convection for both SHCS and the battery provides the most substantial reduction, bringing the temperature down to 37.8°C. A validated computational fluid dynamics (CFD) model using COMSOL Multiphysics is employed to simulate the cooling performance under different boundary Heat pipes, known for their superior thermal conductivity, have become a promising solution for battery cooling applications. This study investigates the combined use of heat pipes and air cooling for LTO battery cells. Experimental data is collected under different cooling conditions and is compared to simulation results from a CFD model.

The cooling methods tested include:- Natural convection without SHCS.- SHCS with natural convection.- SHCS with forced convection.- Forced convection applied to both the SHCS and the battery. The experiments were conducted under controlled conditions to measure the surface temperature of the LTO cells using infrared thermal cameras. The temperature distribution across the cells was recorded, and comparisons were made between the experimental and CFD simulation results. The cooling performance of each method was evaluated based on maximum temperature reduction and uniformity across the battery surface

Liquid Cooling : A liquid cooling system in a Battery Thermal Management System (BTMS) uses a coolant, typically a mixture of water and glycol, to regulate the temperature of lithium-ion batteries. The coolant circulates through a network of pipes or channels adjacent to the battery cells, absorbing heat and transporting it away from the battery. This method is highly effective for high-power applications, as it efficiently dissipates large amounts of heat generated during charging and discharging cycles. Liquid cooling ensures that the batteries operate within an optimal temperature range, enhancing performance and longevity. Although more complex and costly than air cooling systems, liquid cooling provides superior thermal management, making it a preferred choice for high-performance electric vehicles.

LIBs are great for EVs because they store a lot of energy and last a long time, but they need to stay between 15°C and 40°C to work properly. If they get too hot, they can lose capacity or even catch fire. That's why battery thermal management systems (BTMS) are important. Liquid cooling works better than air cooling because it transfers heat more efficiently. This paper reviews how LC-BTMS is used in cylindrical and prismatic batteries, focusing on better designs, working methods, and overcoming challenges like cost and complexity. Liquid Cooling-BTMS is essential to keep EV batteries at safe temperatures. Advanced designs like hybrid cooling systems and better cooling fluids are improving these systems, but they are still expensive and complicated. Cylindrical and prismatic batteries need different solutions because they behave differently when it comes to heat. Numerical simulations help test and improve LC-BTMS designs by studying things like how fast the cooling fluid flows and how well it cools the batteries. The main challenges are to make the systems simpler, cheaper, and more reliable while keeping the batteries at a steady temperature.

Further study in liquid cooling leads to use of much advanced type of liquid such a nano fluid, which can be much better and cost efficient at the same time as Nanofluids have higher thermal conductivity than water, meaning they can transfer heat better.

Results after using Nanofluid as coolant for BTMS :

Nanofluids significantly reduced the battery's maximum temperature compared(by 27.59% in their best module).

Increased Pressure drop: While nanofluids improved cooling the pressure drop inn the cooling system, but this wasn't a major issue.

Conclusions After using Nanofluids as coolant :

Nanofluids are promising coolant for EV batteries de to their superior cooling abilities. This study provides Valuable insights for designing better battery thermal management systems for EVs.

Phase Change Cooling :

Transition of phase of PCM allows it to store more amount of heat in the form of molecular energy called latent heat. PCM are categorized in two classes based on origin one organic and other artificial. Phase Change cooling uses PCM around the source, organic paraffin wax is mostly commonly used PCM remains upto 1500 cycles of phase change

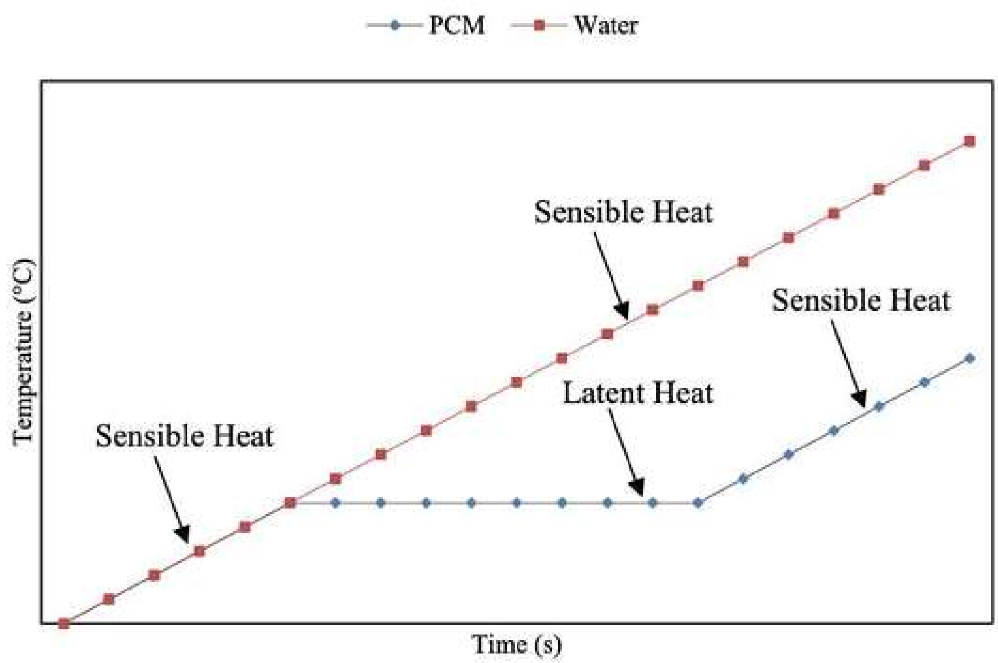

Fig. 1. Thermal energy storage characteristics of PCM.

~ Batteries made using paraffin wax, after discharged and checked for its effectiveness in terms of weight and temp it was found more convenient and below 50°C. Different shapes were also tried. With conventional air current at 0.2m/s rectangular arrangement was found most effective The use of PCM homogenizes the temperature around the battery. A critical limit of thickness of PCM and heat convection coefficient was established through the study(2.08mm,m2K) ~ PCM Fin structure resulted to be superior that PCM Ring structure.

A parametric investigation was conducted on a Passive PCM-Fin BTMS for a Li-Ion battery pack in a high ambient temperature environment. Three PCM layers were adjusted around the battery pack in different thicknesses with 8 fins a ached longitudinally to the battery cells. The maximum temperature difference obtained for high load discharge was 26 K and the internal battery temperature distribution was capped at approximately 5 K.

The cell was then enclosed in an insulated container to discharge using a load bank from 1C and 2.5C. The results indicated that the heat genera on rate increased by approximately 8% as compared to the 1C discharge rate and thus corresponding increment in temperature was recorded. Based on this data, a Paraffin-Graphene CPCM hexagonal structure was con structed to enclose 6 battery cells in a cylindrical package. The resulting combination was found to be effective as a BTMS.

An 18,650 Configuration 3.7 V, 1.2Ah Li-Ion battery cell was experimentally tested by connecting a constant load to it to obtain its temperature and current profiles. Then the battery was wrapped in RT- 47 Paraffin PCM in a 4 mm cavity made by two concentric cylinders making a BTMS.

From the current study, the following conclusions can be derived: 1

. The temperature of the bare battery cell (BA-cell) while discharging grew quasilinear at an average pace of 0.0219 ◦ C/min with a peak of Materials 0.0355 s.

2. The temperature of PCM cooled battery cell (BT-cell) under similar conditions grew almost linearly with a quasi-constant increment of 0.00981 cell. C/min, representing an improvement of 72.366% over BA-

3. The results of the current performance of the BA-cell indicated that it discharged within a dura on of 500 s from star ng current of 4.2A while the BT-cell star ng from 3.7A completed discharge in the 930 s. This represents a performance improvement of 53.763% when compared to the BA-cell.

4. This improvement in battery performance was attributed to be er temperature control as a result of energy absorption by RT-47 Paraffin PCM used in the current BTMS.

5. The obtained results were found to be either in close conformity or be er than previous studies of similar nature..

Without natural convection cooling, cell temperatures reached 69°C at 3C discharge, 60°C at 2C, and 38.7°C at 1C. The study found that PCM cooling significantly reduced cell temperatures, proving most effective without needing external power, thereby helping maintain safe opera ng temperatures for Li-ion cells. Global warming and pollution have pushed researchers to look for cleaner alternatives to fossil fuels. The transportation sector, a major user of conventional resources, is moving towards electric and hybrid electric vehicles (EVs and HEVs) due to their low CO_2 emissions. Li-ion batteries are ideal for these vehicles due to their high capacity, energy density, and efficiency. However, rapid charging and discharging generate significant heat, which can lead to reduced performance, shorter life cycles, and safety risks, including thermal runaway and explosions. The optimal opera ng temperature for Li-ion batteries is between 25°C and 40°C. Effective Battery Thermal Management Systems (BTMS) are essential, and various cooling techniques, including air, water, and phase change materials (PCM), are used to manage battery temperatures. PCM cooling, especially with fa y acid-based PCMs like Lauric acid, is effective in maintaining safe temperatures without external power. Recent studies focus on optimizing these cooling methods to enhance battery safety and performance. This study examines the effects of active and passive cooling methods on the thermal behaviour of a single Li-ion cell and the impact of varying C-rates. Key findings include: Temperature Rise: Without natural convection, the cell's temperature spikes during charging and discharging, risking thermal runaway. Cooling Efficiency: Forced air and PCM cooling significantly reduce cell temperature at higher C-rates compared to natural convection. Forced Cooling: Higher air velocity lowers cell temperature more effectively but increases power consumption. Heat Genera on Factors: Various parameters like C-rate, external temperature, and charging/discharging cycles influence heat genera on and cooling efficiency. PCM Cooling: PCM cooling slightly increases temperature at low C-rates due to PCM's thermal proper es but maintains safe temperatures. Temperature Reductions: At a 3C discharge rate, temperature rises are 39.9°C (natural convection), 20.1°C (forced convection), and 20.8°C (PCM cooling). PCM cooling is most effective at higher C-rates, reducing thermal runaway risks. The study suggests future research should focus on more efficient cooling techniques, be er thermal conductivity PCMs, and higher latent heat for optimal battery thermal management systems.

Summary

Air Cooling

Air cooling involves using airflow—either natural or forced with fans—to dissipate heat from battery cells. It's a cost-effective and simpler solution suitable for moderate temperature management. However, its efficiency is limited in high-power applications due to its lower heat dissipation capacity.

Liquid Cooling

Liquid cooling uses a coolant, typically a water-glycol mixture, circulated through pipes or channels near the battery cells. This method is highly efficient at removing large amounts of heat, making it ideal for high-performance applications. Although more complex and costly, liquid cooling ensures superior thermal regulation, extending battery life and enhancing performance.

PCM (Phase Change Material) Cooling

PCM cooling utilizes materials that absorb and release heat during phase transitions (e.g., from solid to liquid). This method is efficient for maintaining battery temperature without requiring external power. PCMs provide passive cooling by absorbing excess heat, which helps prevent thermal runaway and improves battery safety.

In summary, each cooling method has its unique advantages: air cooling is cost-effective and simple, liquid cooling is highly efficient for high-power needs, and PCM cooling offers passive temperature control without additional power consumption. The choice depends on the specific requirements of the EV and its intended use.

ADVANCED AND HYBRID COOLING

Introduction

Hybrid cooling systems in BTMS are revolutionizing the way we manage battery temperatures in electric vehicles. These systems combine multiple cooling methods to enhance thermal regulation, ensuring that the batteries operate within optimal temperature ranges. This is crucial for maintaining the performance, safety, and longevity of lithium-ion batteries, which are the heart of electric vehicles. In this write-up, we will explore the components, functioning, benefits, and advancements in hybrid cooling systems, focusing particularly on the integration of composite phase change materials (CPCMs) with liquid cooling and nanoadditives.

Importance of BTMS in EVs

BTMS are essential for electric vehicles as they regulate the temperature of the battery packs, preventing overheating and ensuring efficient operation. The rapid charge and discharge cycles in EV batteries generate significant heat, which, if not managed properly, can lead to reduced performance, shorter battery life, and safety risks such as thermal runaway. Traditional cooling methods like air and liquid cooling have their own limitations, which hybrid systems aim to overcome.

Components of Hybrid Cooling Systems

Composite Phase Change Materials (CPCMs)

Phase change materials (PCMs) are substances that absorb and release heat during phase transitions, such as from solid to liquid. CPCMs are enhanced PCMs that incorporate materials like expanded graphite or nanoadditives to improve their thermal conductivity. These materials are highly effective in managing temperature fluctuations without requiring external power.

Liquid Cooling

Liquid cooling systems use a coolant, typically a mixture of water and glycol, circulated through pipes or channels adjacent to the battery cells. The coolant absorbs the heat generated by the batteries and transports it away, effectively dissipating large amounts of heat. This method is more efficient than air cooling, especially in high-power applications.

Nanoadditives

Nanoadditives are materials at the nanoscale that, when added to coolants or PCMs, enhance their thermal properties. These additives can significantly improve the thermal conductivity and heat absorption capacity of the cooling materials, making the hybrid system more efficient.

How Hybrid Cooling Systems Work

Hybrid cooling systems combine the strengths of active and passive cooling methods to provide comprehensive thermal management. Here's how they function:

1. Heat Absorption: During charging and discharging, the battery cells generate heat. The CPCMs, integrated within the battery pack, absorb this heat during their phase transition, effectively regulating the temperature.

2. Heat Dissipation: The liquid cooling system circulates the coolant through the battery pack, absorbing the heat from the CPCMs and the battery cells. The heated coolant is then transported away from the battery pack to a radiator or heat exchanger, where the heat is dissipated into the environment.

3. Enhanced Conductivity: Nano additives in the CPCMs and the coolant enhance their thermal conductivity, ensuring quicker and more efficient heat transfer. This helps in maintaining the battery temperature within the optimal range, even during high-demand scenarios like fast charging or high-power discharges.

Benefits of Hybrid Cooling Systems

Improved Efficiency

Hybrid systems provide superior cooling efficiency compared to traditional methods. By combining the passive cooling of CPCMs with the active cooling of liquid systems, these systems can manage higher heat loads effectively.

Enhanced Safety

By preventing overheating and managing thermal runaway risks, hybrid cooling systems significantly enhance the safety of electric vehicles. The use of CPCMs, which do not require external power to function, adds an extra layer of reliability.

Extended Battery Life

Effective thermal management reduces the stress on battery cells, slowing down degradation and extending their lifespan. This means fewer replacements and lower maintenance costs over time.

Consistent Performance

Maintaining the battery temperature within the optimal range ensures consistent performance. Whether it’s acceleration, range, or charging speed, the vehicle performs reliably under various conditions.

Advancements in Hybrid Cooling Systems

Recent research and technological advancements are pushing the boundaries of hybrid cooling systems. Here are some notable developments:

Advanced CPCMs

Researchers are developing CPCMs with enhanced thermal properties using materials like graphene and carbon nanotubes. These materials offer superior thermal conductivity and heat absorption capacity, making the cooling systems more efficient.

Smart BTMS

Integration of smart sensors and control systems in BTMS allows real-time monitoring and management of battery temperatures. These systems can adjust the cooling mechanisms based on the battery’s state, ensuring optimal performance and safety.

Nano additives

The use of nano additives is continuously being optimized. New nano additives are being explored for their potential to further enhance the thermal properties of CPCMs and coolants, pushing the efficiency of hybrid cooling systems to new heights.

Case Studies

Case Study 1: Performance of Hybrid Cooling in High-Power EVs

A study conducted on a high-power electric vehicle equipped with a hybrid cooling system demonstrated a significant reduction in battery temperature. The integration of CPCMs with liquid cooling and nano additives reduced the peak temperature by up to 50% compared to traditional cooling methods. This not only improved the vehicle's performance but also extended the battery life by reducing thermal stress.

Case Study 2: Cost-Benefit Analysis

An analysis of the cost-benefit ratio of hybrid cooling systems in EVs showed that while the initial investment is higher, the long-term benefits in terms of extended battery life and enhanced safety far outweigh the costs. The reduced need for battery replacements and the prevention of thermal runaway incidents contribute to overall cost savings and increased vehicle reliability.

Conclusion

Hybrid cooling systems in BTMS represent a significant advancement in the thermal management of electric vehicles. By combining the strengths of composite phase change materials, liquid cooling, and nano additives, these systems offer superior temperature regulation, enhanced safety, and improved battery performance. The ongoing research and development in this field promise even more efficient and effective cooling solutions in the future, paving the way for more reliable and high-performing electric vehicles. As the EV industry continues to grow, the importance of robust BTMS cannot be overstated, and hybrid cooling systems are at the forefront of this technological evolution.